

IMS22IS138

Programme : B E – Information Science & Engineering

Internal Assessment – I

TERM : 15 th April to 27 th July 2024	COURSE NAME: Microcontrollers
DATE : 30/05/2024 TIME: 9.30am-10.30pm	COURSE CODE : IS42
MAX MARKS: 30	PORTIONS : L1 - L20



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO's	Marks
1 a)	Discuss the purpose of using the pulse width modulator in MCU's. Show how to achieve 50% duty cycle in PWM.	L3	CO1	07
b)	With the timing diagram, Explain asynchronous memory read cycle of DRAM chip.	L2	CO2	08
2 a)	What is a reset? List and explain the types of reset in a typical MCU.	L2	CO1	08
b)	List and explain the steps involved in I2C on board protocol and write the positive aspects of I2C protocol.	L3	CO2	07
3 a)	Discuss with a neat diagram, the internal components of a typical MCU that works on System on Chip.	L2	CO1	07
b)	i) Write the following numbers in the big endian and little endian format in the address 0x450011: i) 0xDC67 ii) 0x1908DEF8. ii) Consider the following example: LDR r1, =0xffffffff LDR r2, =0x00000001 ADDS r0, r1, r2 Identify how different types of flags will be set.	L2	CO2	08

*L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create